

Jeslin P James

✉ jeslin@ug.cusat.ac.in | 🌐 jeslinpjames.in | 📧 jeslin-p-james | 📞 jeslinpjames | 📞 +91 9383468783

EDUCATION

Cochin University of Science and Technology

Kochi, India

Five Year Integrated M.Sc. in Computer Science (AI & Data Science)

Saint Francis School

Vadakkencherry

ISC (Class 12, 2021) — ICSE (Class 10, 2019) – Computer Science

EXPERIENCE

AI & Automation Engineer (Part-time)

Apr 2025 – Present

Semicon Fab Technologies Private Limited

- Built a cleanroom particle-control system that tracks human motion (the primary particle source) and predicts future contamination spikes, triggering reactive HVAC responses to maintain air cleanliness.
- Engineered an **LSTM+Attention** forecasting model that predicts particulate concentrations per zone from live sensor streams, with automated online retraining and outlier-robust scaling directly against production **InfluxDB** data.
- Designed a predictive energy-control loop that maps forecasted particle load to HVAC fan speed via **Modbus TCP** PLC integration, computing real-time power and CO₂ savings against a reactive baseline.
- Architected and deployed a containerized IoT stack (**Docker**, **Node-RED**, **InfluxDB**, **Grafana**) with live dashboards for particle counts, occupancy, and HVAC metrics, hardened as fault-tolerant **systemd** services with automatic recovery.

AI & Backend Engineer Intern

Aug 2025 – Mar 2026

Laennec AI

- Lead AI research for respiratory sound classification, architecting a CNN+BiLSTM+Attention pipeline in PyTorch; achieved 98% accuracy (up from 86%) via custom feature engineering and K-fold ensembling.
- Own and manage cloud infrastructure on **Google Cloud Platform (GCP)**, configuring GCS buckets and Compute Engine instances for high-performance model training and scalable deployment.
- Develop and maintain robust backends using FastAPI to serve model inferences, ensuring low-latency communication between the AI core and the application layer.

Backend & AI/ML Developer (Contract)

Nov 2024 – Apr 2025

Provez, Dubai / Remote

- MVP *Legal Case Manager*: case/matter CRUD, doc ingest/search (PDF, Word, scans), calendar, LLM drafting/summaries; multi-tenant ready.
- Owned backend: FastAPI, Postgres/Alembic, JWT auth, Docker; CI (pytest+GitHub Actions).

Backend & AI/ML Development Intern

Aug 2024 – Nov 2024

GreenFI AI (ESG / Carbon Analytics), Singapore / Remote

- ESG emissions pipeline: ingest multilingual utility bills (PDF/img/CSV) → OCR/LLM parse (OpenAI) → normalized energy for carbon calc.
- Asset linking via embedding search (pgvector/Postgres); batch ZIP upload; Flask API on AWS EC2; cut manual asset mapping ~90% on pilot (500 bills).

Student Research Intern

May 2024 – Jul 2024

Indian Institute of Space Science & Technology (IIST), Thiruvananthapuram

- Early crop pest detection: built synthetic data pipeline (21K images + 1K real, 11 pest classes) via custom compositing tuned for small objects.
- YOLO variants & FocusDET slice proto; Darknet/CUDA C++ training (~6× faster vs PyTorch proto) and improved small-object detection in field samples.

Open Source Contributions

2024

- 📦 **roboflow/supervision** (computer vision utilities)
- 📦 **hpcaitech/Open-Sora** (video generative AI)

PUBLICATIONS

Explainable Raga Classification Under Extreme Data Scarcity: A Branch-wise Ablation and Multi-Modal Explainability Study – Jeslin P. James, Shailesh S. *Big Data Analytics (BDA 2026)*, BITS Pilani, Goa Campus, India.

PROJECTS

Multi-Class Object Localization with Deep RL – Developed a GRU-based agent using Grad-CAM and 6 dynamic actions to localize objects across multiple classes.

Statistical Image Synthesis via Matrix Variate Distributions – Developed a framework using pixel-wise mean, variance and neighborhood covariance (matrix variate normal) to directly sample and generate high-fidelity grayscale images; validated on MNIST and Fashion-MNIST with strong SSIM scores.

Share More: P2P File Sharing and Watch Party Platform - Developed a platform for P2P file sharing and real-time watch parties. GitHub

StudySphere: Personalized Learning Platform - Built a platform using React and Django to assist students with notes, flashcards, and quizzes.

Kathakali Navarasa Classification Using Vision Transformer (ViT) and LSTM - Developed a model to classify traditional Kathakali expressions from video data.

Implemented Transformer for Malayalam Language Model - Fine-tuned a Transformer model for the Malayalam language.

TECHNICAL SKILLS

Languages: Python, C++, C, SQL Java, JavaScript, R, LaTeX, Golang

ML/CV/NLP: PyTorch, TensorFlow, OpenCV, CUDA, HuggingFace Transformers

Backend & Data: FastAPI, Flask, Postgres (pgvector), Alembic, Docker, GitHub Actions

Cloud/LLM: AWS (EC2/S3), GCP, OpenAI API